

Topic 2.5 Natural language to FOL Conversion, and application of the Resolution rule

- a) Resolution requires, the KB is in CNF, i.e. each sentence is a clause or disjunction of FOL literals, and the clauses are in conjunction to each other.

We show the conversion in an example, and then apply Resolution to answer a query.

b) Example of conversion

i) KB in Natural Language:

1. Every guest receives at least one gift.
2. Karim is a guest.

ii) KB in FOL :

1. $\forall x (\text{Guest}(x) \Rightarrow \exists y (\text{Gift}(y) \wedge \text{Receives}(x, y)))$
2. $\text{Guest}(\text{Karim})$

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2. $\text{Guest}(\text{Karim})$

iii) Conversion to CNF:

- I. Standardize variables. [Use separate variables for different quantifiers.]
- II. Eliminate $\Rightarrow$ and $\Leftrightarrow$, and move $\neg$ inward.

$$\forall x (\neg \text{Guest}(x) \vee (\exists y (\text{Gift}(y) \wedge \text{Receives}(x, y))))$$
- III. Skolemize

$$\forall x (\neg \text{Guest}(x) \vee (\text{Gift}(\text{GiftFor}(x)) \wedge \text{Receives}(x, \text{GiftFor}(x))))$$
- IV. Drop all $\forall$ s.

$$(\neg \text{Guest}(x) \vee (\text{Gift}(\text{GiftFor}(x)) \wedge \text{Receives}(x, \text{GiftFor}(x))))$$
- V. Simplify further to get CNF.

$$(\neg \text{Guest}(x) \vee \text{Gift}(\text{GiftFor}(x))) \wedge$$

$$(\neg \text{Guest}(x) \vee \text{Receives}(x, \text{GiftFor}(x)))$$

iv) KB in CNF of FOL:

1. $\neg \text{Guest}(x) \vee \text{Gift}(\text{GiftFor}(x))$
2. $\neg \text{Guest}(x) \vee \text{Receives}(x, \text{GiftFor}(x))$
3. $\text{Guest}(\text{Karim})$

iv) KB in CNF of FOL:

1. $\neg \text{Guest}(x) \vee \text{Gift}(\text{GiftFor}(x))$
2. $\neg \text{Guest}(x) \vee \text{Receives}(x, \text{GiftFor}(x))$
3. $\text{Guest}(\text{Karim})$

d) Query: Does Karim receive a gift?

$\exists x (\text{Gift}(x) \wedge \text{Receives}(\text{Karim}, x))$

i) To answer / prove, add the negation of the sentence to the KB and try to derive a contradiction.

ii) Equivalently,

Add the CNF of the negation of the query to the KB, and try to derive an empty clause ($[]$) by applying the Resolution rule repeatedly.

iii) If $[]$ can't be derived, it means that the sentence (query) is False.
This is known as the Resolution-Refutation completeness of KB.

e) Conversion of the negation of the query:

i) $\neg(\exists x (\text{Gift}(x) \wedge \text{Receives}(\text{Karim}, x)))$

ii) $\forall x \neg(\text{Gift}(x) \wedge \text{Receives}(\text{Karim}, x))$

iii) $\forall x (\neg\text{Gift}(x) \vee \neg\text{Receives}(\text{Karim}, x))$

iv) $\neg\text{Gift}(x) \vee \neg\text{Receives}(\text{Karim}, x)$

f) The KB after adding the negation of the query:

1. $\neg\text{Guest}(x) \vee \text{Gift}(\text{GiftFor}(x))$

2. $\neg\text{Guest}(x) \vee \text{Receives}(x, \text{GiftFor}(x))$

3. $\text{Guest}(\text{Karim})$

T1. $\neg\text{Gift}(y) \vee \neg\text{Receives}(\text{Karim}, y)$

KB: 1. $\neg \text{Guest}(x) \vee \text{Gift}(\text{GiftFor}(x))$
 2. $\neg \text{Guest}(x) \vee \text{Receives}(x, \text{GiftFor}(x))$
 3. $\text{Guest}(\text{Karim})$
 T1. $\neg \text{Gift}(y) \vee \neg \text{Receives}(\text{Karim}, y)$

g) Finding the answer to the query :

I. Resolving T1 and 1 with $\theta = \{y / \text{GiftFor}(x)\}$:

T2. $\neg \text{Guest}(x) \vee \neg \text{Receives}(\text{Karim}, \text{GiftFor}(x))$

II. Resolving T1 and 2 with $\theta = \{x / \text{Karim}, y / \text{GiftFor}(\text{Karim})\}$:

T3. $\neg \text{Guest}(\text{Karim}) \vee \neg \text{Gift}(\text{GiftFor}(\text{Karim}))$

III. Resolving T2 and 2 with $\theta = \{x / \text{Karim}\}$: T4. $\neg \text{Guest}(\text{Karim})$

IV. Resolving T2 and 3 with $\theta = \{x / \text{Karim}\}$:

T5. $\neg \text{Receives}(\text{Karim}, \text{GiftFor}(\text{Karim}))$

~~V. Resolving T3 and 1 with $\theta = \{x / \text{Karim}\}$: T6. $\neg \text{Guest}(\text{Karim})$~~

VI. Resolving T3 and 3 with $\theta = \{\}$: T7. $\neg \text{Gift}(\text{GiftFor}(\text{Karim}))$

VII. Resolving T4 and 3 with $\theta = \{\}$: []

As an **empty clause** is resolved, the sentence (query) is proved true, that is, the answer is 'Yes, Karim receives a gift.'